



Microbiology of the Oral Cavity — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Approximately how many bacterial species inhabit the oral cavity?

EASY

- A. About 20
- B. Over 700
- C. About 70
- D. About 7,000
- E. Fewer than 10

2. Which genus is the most abundant at every oral niche shown in the lecture figure?

EASY

- A. *Prevotella*
- B. *Fusobacterium*
- C. *Neisseria*
- D. *Streptococcus*
- E. *Gemella*

3. Which event creates the surface on which dental plaque accumulates?

EASY

- A. Weaning
- B. Puberty
- C. Tooth eruption
- D. Antibiotic use
- E. Salivary flow

4. Which is an antimicrobial component of saliva?

EASY

- A. Pepsin
- B. Gastrin
- C. Bile salts
- D. Insulin
- E. Secretory IgA

5. Microbes acquired from food, hands and close contacts and usually removed by saliva flow are called:

MEDIUM

- A. Core microbiota
- B. Resident microbiota
- C. Commensal biofilm
- D. Transient microbiota
- E. Variable core

6. Which oral site has the lowest oxygen availability?

MEDIUM

- A. Exposed tooth surfaces
- B. Tongue dorsum tip
- C. Gingival crevices
- D. Lips
- E. Hard palate

7. What is the first step in dental plaque formation?

MEDIUM

- A. Co-aggregation
- B. Dysbiosis
- C. Acquired pellicle
- D. Matrix maturation
- E. Streptococcal adhesion

8. A dentist explains that caries reflects the biofilm environment. Which pair is most associated with acid production?

MEDIUM

- A. *Porphyromonas* and *Capnocytophaga*
- B. *Veillonella* and *Neisseria*
- C. *Candida* and *Actinomyces*
- D. *Fusobacterium* alone
- E. *S. mutans* group and *Lactobacillus*

9. A patient develops subacute endocarditis after a dental extraction. Which organisms are classic?

MEDIUM

- A. *S. pyogenes*
- B. *C. diphtheriae*
- C. *N. gonorrhoeae*
- D. Viridans streptococci
- E. *Candida*

10. Which condition makes oral *Candida* and invasive fungal lesions more clinically important?

MEDIUM

- A. Recent tooth eruption
- B. Vigorous brushing
- C. High salivary flow
- D. Immunocompromised host
- E. Vegetarian diet

11. A pharyngitis complication is immune-mediated and involves heart, joints, skin and CNS. Name it.

MEDIUM

- A. Acute rheumatic fever
- B. Acute glomerulonephritis
- C. Peritonsillar abscess
- D. Deep neck infection
- E. Diphtheria

12. Which set of findings matches acute glomerulonephritis after streptococcal infection?

MEDIUM

- A. Oedema, hypertension, haematuria, proteinuria
- B. Carditis and chorea
- C. Bull neck and membrane
- D. Koplik spots
- E. White removable plaques

13. Which finding is characteristic of diphtheria?

MEDIUM

- A. Painful vesicles
- B. Adherent gray pharyngeal pseudomembrane
- C. Koplik spots
- D. White plaques that scrape off
- E. Silent asymptomatic infection

14. Koplik spots on the buccal mucosa suggest:

MEDIUM

- A. HSV infection
- B. Measles
- C. Oral thrush
- D. Diphtheria
- E. Gonococcal pharyngitis

15. Two children have white oral coating. In one it wipes off leaving a red surface; in the other the gray membrane is adherent. What are they, respectively?

HARD

- A. Diphtheria; thrush
- B. Thrush; diphtheria
- C. Thrush; measles
- D. Measles; herpes
- E. Herpes; diphtheria

16. Why does broad-spectrum antibiotic use predispose to oral thrush?

HARD

- A. It disturbs the resident microbiota and the balance that limits *Candida*
- B. It raises salivary pH permanently
- C. It removes salivary IgA
- D. It sterilises the mouth
- E. It reduces host immunity to viruses

17. A patient with xerostomia has recurrent oral infections. Which mechanism best explains this?

HARD

- A. Excess salivary IgA
- B. Higher oxygen in the crevice
- C. Increased tooth eruption
- D. Loss of transient microbiota
- E. Loss of saliva flow reduces removal of microbes and delivery of IgA and lysozyme

18. Two patients eat the same total sugar: one in a single sitting, the other sipping through the day. Which has the higher caries risk and why?

HARD

- A. The sipper — repeated fermentation keeps plaque pH low
- B. The single-sitting patient — more sugar per event
- C. Neither — sugar does not affect plaque
- D. The sipper — higher salivary IgA
- E. The single-sitting patient — more *Lactobacillus*

19. A sexually active 21-year-old has a mild sore throat and no other symptoms. Which statement is most appropriate?

HARD

- A. Exclude gonorrhoea because it does not affect the pharynx
- B. Treat as diphtheria
- C. Consider gonococcal pharyngitis and request throat-site testing
- D. Assume viral and require no testing
- E. It is only seen in immunocompromised patients

20. A child with a skin infection develops oedema, hypertension and dark urine two weeks later. Which is the best interpretation?

HARD

- A. Acute rheumatic fever
- B. Peritonsillar abscess
- C. Diphtheria toxin effect
- D. Thrush
- E. Acute glomerulonephritis after streptococcal skin infection

✓ Answer key

Q	Answer	Difficulty	Why
1	B	EASY	The lecture states over 700 bacterial species plus fungi, protozoa, archaea and viruses.
2	D	EASY	Streptococcus is red (most abundant) at all nine sites.
3	C	EASY	Teeth are a permanent, non-shedding surface.
4	E	EASY	Saliva contains secretory IgA, lysozyme, peroxidase and antimicrobial peptides.
5	D	MEDIUM	Transient microbiota are removed by competitors, immune defences and saliva; they may cause infection if barriers fail.
6	C	MEDIUM	O ₂ is highest at exposed surfaces and lowest in gingival crevices.
7	C	MEDIUM	Steps: pellicle → early streptococcal adhesion → co-aggregation → maturation + matrix → dysbiosis.
8	E	MEDIUM	S. mutans is the major caries-associated group; Lactobacillus adds acid production in established acidic environments.
9	D	MEDIUM	Dental procedures cause transient bacteraemia; viridans streptococci are the classic oral cause.
10	D	MEDIUM	Immunosuppression raises the importance of oral Candida.
11	A	MEDIUM	Timely treatment of GAS pharyngitis reduces the risk of rheumatic fever.
12	A	MEDIUM	AGN is immune-mediated and may follow pharyngeal or skin infection.
13	B	MEDIUM	Toxin-producing C. diphtheriae causes an adherent membrane with systemic toxicity.
14	B	MEDIUM	Koplik spots are the oral sign of measles.
15	B	HARD	Thrush plaques often can be removed; the diphtheritic pseudomembrane is adherent.
16	A	HARD	Antibiotics and steroids disturb balance; antibiotics are a listed risk for thrush.
17	E	HARD	Saliva dilutes and removes microbes and carries antimicrobial factors; xerostomia is a listed risk factor.
18	A	HARD	Diet frequency matters: repeated pH drops favour acidogenic biofilm and demineralisation.
19	C	HARD	It may be mild or asymptomatic and needs site-appropriate testing.
20	E	HARD	AGN may follow pharyngeal or skin infection, whereas rheumatic fever follows pharyngitis.